MATHS / ALGEBRA / FACTORISING

Teacher solutions — verified by expansion

PAGE 1 · HIGHEST COMMON FACTOR

EVENS: CLASS / ODDS: HOME

Numerical method: divide every term by the same HCF.

HCF — two algebraic terms

Example: 12x + 18 = 6(2x + 3)

1. 8x + 12
   = 4(2x + 3)

2. $12x^2 + 18x$
   = 6x(2x + 3)

3. $14xy - 21x$
   = 7x(2y − 3)

4. $18x^3y^2 + 24x^2y$
   = $6x^2y(3xy + 4)$

5. $20a^2b + 30ab^2$
   = 10ab(2a + 3b)

6. $24x^4y^3 - 36x^2y^2$
   = $12x^2y^2(2x^2y - 3)$

7. $35m^3n + 49m^2n^2$
   = $7m^2n(5m + 7n)$

8. $30x^5y^3z - 45x^4y^2z^2$
   = $15x^4y^2z(2xy - 3z)$

HCF — three algebraic terms

Example: 6x + 12y + 18 = 6(x + 2y + 3)

1. 6x + 12y + 18
   = 6(x + 2y + 3)

2. $12x^2 + 18x + 30x^3$
   = $6x(2x + 3 + 5x^2)$

3. 15xy + 20xz − 25x
   = 5x(3y + 4z − 5)

4. $18x^3y^2 + 24x^2y^3 - 30x^2y^2$
   = $6x^2y^2(3x + 4y - 5)$

5. $21p^2q + 28pq^2 + 35pq$
   = 7pq(3p + 4q + 5)

6. $24a^3b^2 - 36a^2b^3 + 48a^2b^2$
   = $12a^2b^2(2a - 3b + 4)$

7. $16x^5y^2 + 24x^3y^3 - 40x^4y^2$
   = $8x^3y^2(2x^2 + 3y - 5x)$

8. $-30x^5y^3z + 45x^4y^2z^2 + 15x^3y^4z$
   = $15x^3y^2z(-2x^2y + 3xz + y^2)$

Check by multiplying out the brackets.

A negative factor reverses both signs.

PAGE 2 · FACTORISING BY GROUPING

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Make the two brackets identical; sometimes take out a negative.

Grouping — matching brackets

Example: xy + 2y + xz + 2z = (x + 2)(y + z)

1. xy + xz + 2y + 2z
   = (x + 2)(y + z)
2. $2x^2 + 6xy + 3xz + 9yz$
   = (x + 3y)(2x + 3z)
3. ax + ay + bx + by
   = (a + b)(x + y)
4. $6x^2 + 9xy + 4xz + 6yz$
   = (2x + 3y)(3x + 2z)
5. $x^3 + 2x^2 + 3x + 6$
   = (x + 2)(x² + 3)
6. $4x^4y + 8x^3y^2 + 3x^2z + 6xyz$
   = (x + 2y)(4x³y + 3xz)
7. $a^2b + 3ab^2 + 2ac + 6bc$
   = (a + 3b)(ab + 2c)
8. $10x^5y^2 + 15x^4y^3 + 6x^2z + 9xyz$
   = (2x + 3y)(5x⁴y² + 3xz)

Grouping — take out a negative

Example: x(y−z) + 3(z−y) = (x−3)(y−z)

1. xy − xz + 3z − 3y
   = (x − 3)(y − z)
2. $x^2 - xy - 5x + 5y$
   = (x − 5)(x − y)
3. $a^2 + 2ab - 3ac - 6bc$
   = (a + 2b)(a − 3c)
4. $4x^3 - 8x^2y - 3xz + 6yz$
   = (x − 2y)(4x² − 3z)
5. 2ab − 4ac + 6c − 3b
   = (2a − 3)(b − 2c)
6. $6x^4y^2 - 9x^3y^3 - 4xz + 6yz$
   = (2x − 3y)(3x³y² − 2z)
7. $x^5 - x^4y - 7x + 7y$
   = (x − y)(x⁴ − 7)
8. $12a^5b^2 - 18a^4b^3 - 20a^2c + 30abc$
   = 2a(2a − 3b)(3a³b² − 5c)

Check by multiplying out the brackets.

A negative factor reverses both signs.